LEXON

Workspace + Team System & Setup Guide

Supabase setup, SQL, login (Google + password), Workspace-Team connection, YouTube, Office/API system, data map, Windows + web. Language: Hinglish (technical terms English). Version: October 2026.

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Pehle ye padho: Ye guide aapke diye hue files (lexon-workspace.zip, lexon-team.zip, LEXON_A_workspace_team.sql) ko padh kar bani hai. Office app, aur YouTube/Instagram/Office ke alag SQL files (LEXON_B/C/D) mujhe nahi mile, isliye unke andar ka code maine check nahi kiya. Un jagah sirf wahi likha hai jo Workspace code aur purani docs se confirm hua.

1. Architecture: 3 Supabase projects, 2 apps

Final structure (Workspace code, gateway function aur docs se confirm):

Project	Kaun use karta hai	Kya rakhta hai
A (shared)	Lexon Workspace (admin) + Lexon Team (members) dono	Auth, team, tasks, projects, messages, files, notifications, calls, integrations (keys Vault me), automations, agents, activity
B Office	Office app (alag virtual environment)	Office members, agents, layout, events. Workspace ke real team members yahan copy nahi hote
C+D Automation (ek hi project)	YouTube + Instagram automation, ek URL aur ek service key	YouTube: Google tokens, snapshots, videos, AI outputs. Instagram: rules, events, actions, messages, analytics, logs. Sirf service role, koi client policy nahi

Workspace B aur Automation project tak **sirf** `project-gateway` Edge Function se pahunchta hai: login/permission check, whitelisted tables, rate limit. B aur Automation project ki service key kabhi browser me nahi aati, Vault me encrypted rehti hai.

```
Browser / Windows app → Supabase A (Auth + RLS) → Edge Functions → B, C, D
```

2. Supabase kaise setup aur manage hoga

2.1 Project A banao

1. supabase.com par naya project (region Mumbai). Database password safe jagah save karo (kisi file me commit nahi).
2. Settings → API: **Project URL** aur **anon key** copy karo. Ye public values hain. **service_role key kahin paste mat karo** (frontend, chat, GitHub).
3. SQL Editor me `LEXON_A_workspace_team.sql` paste karke Run (section 4).
4. Authentication settings (section 5.4): Google ON, Email ON, password min length 8, redirect URLs.
5. Database → Extensions: `pg_cron` ON (7-din cleanup raat ko chalta hai; na ho to app admin ke khulne par roz ek baar chalata hai).
6. Edge Functions deploy (section 2.3) aur secrets set karo.
7. **Sabse pehle khud Workspace me Google se login karo.** Pehla account Owner/Admin ban jata hai. Uske baad sirf approved emails andar ja sakte hain.

2.2 Dono apps me same values

```
apps/web/.env    (Workspace)     apps/team/.env   (Team)
VITE_SUPABASE_URL=https://<A-ref>.supabase.co     <- dono me SAME
VITE_SUPABASE_ANON_KEY=<A anon key>               <- dono me SAME
```

2.3 Edge Functions (project A)

```
supabase login
supabase link --project-ref <A-ref>
supabase functions deploy integrations-oauth --no-verify-jwt
supabase functions deploy project-gateway
supabase functions deploy invite-member
supabase functions deploy ai-keys
supabase functions deploy assistant
supabase functions deploy office-ai --no-verify-jwt      # Office Workers ke liye (section 7)
cp supabase/.env.example supabase/.env      # values bharo (git me commit nahi)
supabase secrets set --env-file supabase/.env
```

Secret	Kis liye
ALLOWED_ORIGINS	Workspace URL, Team URL, lexon://app, lexon-team://app (comma se alag). Kabhi * nahi
APP_URL, TEAM_APP_URL	Redirects aur invite links
OAUTH_STATE_SECRET	Google/GitHub OAuth state sign karne ke liye (random lamba string)
GOOGLE_CLIENT_ID / SECRET	Integrations (Drive, Calendar, Gmail) ke Google OAuth
OFFICE_GATEWAY_SECRET	Office server → office-ai function ka shared secret (section 7)

2.4 Roz ka management

- **Data dekhna/theek karna:** Supabase → Table Editor. **Users:** Authentication → Users.
- **Naya member:** Workspace → Team → Invite (invite-member function). **Naya Admin:** Settings → Admin access (sirf Owner).
- **Health check:** SQL Editor me `supabase/DIAGNOSE.sql` (sirf padhta hai) chalao. Jo row ok = false hai wahi missing hai.
- **Backup:** Supabase paid plan me daily backups; free plan par kabhi-kabhi SQL dump/export lo.

3. Workspace aur Team kaise connected hain

- **Same database, same login:** dono apps ek hi project A ka URL aur anon key use karte hain. Alag sync ya copy nahi, ek hi tables.
- **Same RLS:** permissions database me enforce hote hain (has_permission, is_admin, can_see_task). Admin ko sab; Team Member ko fixed access: apne tasks, apni activity, apni chats.
- **Realtime:** Workspace me task assign karo to Team me notification turant aati hai (triggers + Realtime). Messages aur calls dono apps me same conversation dikhate hain.
- **Ek hi SQL:** Team ka apna koi SQL nahi hai. Team jo tables/RPCs use karta hai (profiles, tasks, messages, notifications, call_sessions, my_conversations, start_direct, create_group, toggle_reaction, set_security_retention ...) sab LEXON_A file me hain (section 11).
- **Same look:** Team Workspace jaisi fonts (Inter), colors, sidebar, header aur cards use karta hai, upar se halke rounded corners aur mobile par iPhone-style bottom bar.

Kaam	Workspace (admin)	Team (member)
Login	Google ya email+password	Google, email+password, ya email code
Dikhta kya hai	Home, Messages, Office, Projects, Team, Agents, Tasks, Files, Analytics, Automations, Integrations, Settings	Home, Messages, Activity, Notifications, Settings
Member ka data	Sab dekh sakta hai	Sirf apna (tasks, activity); chats sirf jisme member ho

4. SQL kahan aur kaise run karna hai

Ek hi file: `supabase/LEXON_A_workspace_team.sql` (Workspace aur Team dono ke liye). Ye migrations 0001 se 0008 + 0009 (repair) + 0010 (hardening) ko ek me jodti hai. Original migrations `supabase/migrations/` me bhi hain.

Naya Supabase project (khali)

1. Supabase Dashboard → **SQL Editor** → New query.
2. Poori `LEXON_A_workspace_team.sql` paste karo → **Run** (sirf ek baar).
3. Error aaye to line number dekho. Schema-cache error ho to chalao: `notify pgrst, 'reload schema';`
4. `DIAGNOSE.sql` chalao: koi ok = false nahi hona chahiye.

Purana project (pehle se SQL chala hua)

Poori file dobara run mat karo (enums/tables dobara banane par error aayega). Sirf wo migrations chalao jo missing hain: 0008, phir 0009, phir 0010. Pata karne ke liye DIAGNOSE.sql chalao.

Migration	Kya karta hai	Kab chalana
0008_single_admin_v2	Roles sirf Admin + Team Member, projects v2, retention, invite lifecycle, profiles.assistant_name	Agar 0001-0007 tak hi chala tha
0009_home_repair	Missing assistant_name/description columns add karta hai. Dobara chalana safe	Home "Could not load data" dikhaye
0010_auth_hardening	Deactivate kiye member ko chats se bhi bahar karta hai	Hamesha (safe)

Password SQL me nahi hota. Supabase Auth sirf bcrypt hash `auth.users` me rakhta hai. Hamare kisi table me password ya hash nahi hai.

5. Authentication kaise kaam karega

5.1 Teen tareeke, ek hi allow-list

Tareeka	Kahan	Kaise verify hota hai
Google	Workspace + Team	Supabase Auth Google OAuth. Naya account banta hai to database trigger `handle_new_user` email ko `authorized_emails` me dhundta hai; nahi mila to **error, account banta hi nahi**
Email + password (8+ chars)	Workspace + Team	`supabase.auth.signInWithPassword`. Supabase hash compare karta hai. Galat email aur galat password dono par ek hi message ("Invalid email or password")
Email code (6 digit)	Team (pehli baar / password bhoolne par)	Invite ke baad ya forgot password me. Login ke baad Settings → Security me password set karo

Password set kaise hota hai: naya member invite code ya Google se pehli baar login karta hai, phir **Settings → Security → Password login** me 8+ character ka password set karta hai. Password sirf TLS par Supabase Auth ko jata hai. Admin kisi ka password dekh ya set nahi kar sakta.

5.2 Sirf authorized Gmail

- Gate **database me** hai, sirf screen par nahi. Koi anon key ke saath direct signUp/OAuth bhi try kare to trigger reject karta hai.
- Pehla user (jab koi workspace nahi) Owner banta hai. Uske baad naye Admin sirf Owner approve kar sakta hai (Settings → Admin access).
- Member ko invite: Team → Invite. Email `authorized_emails` me jata hai (role hamesha member).

5.3 Aur suraksha

- **2FA (TOTP):** Settings → Security. ON hone par database bhi bina aal2 token ke data nahi deta (restrictive policy `mfa_gate`).
- **Idle logout:** 30 minute inactivity par auto sign-out. **Sessions:** "Sign out other devices" / "everywhere".
- **Deactivated member:** app me "Access removed" screen aati hai aur database har policy me block karta hai (0010 se chats bhi).
- **Login history** security_events me (7-30 din, setting).

5.4 Supabase Dashboard me ye settings zaroor karo

1. Authentication → Sign In / Providers → **Email: ON**. **Minimum password length = 8** (server par enforce hoga; app me bhi check hai).
2. Providers → **Google: ON** (Google Cloud me OAuth client banao; redirect URI = Supabase callback URL).
3. URL Configuration: **Site URL** = Workspace URL. **Redirect URLs** me daalo: Workspace URL, Team URL, `http://localhost:5173/**`, `lexon://app/**`, `lexon-team://app/**`.
4. Email Templates → Magic Link me `{{ .Token }}` zaroor rakho (Team ka 6-digit code).
5. SMTP: apna SMTP lagao (Resend/Brevo). Default Supabase email bahut limited hai.
6. Multi-Factor: TOTP ON. Rate limits dekh lo.

Forgot password: Team me "Get an email code" se login karke Settings me naya password set karo. Workspace me Google se login karo (ya Owner se dobara approve karwao). Alag "reset email link" flow abhi nahi banaya.

6. YouTube integration: kahan configure hogi

YouTube ka **saara data aur Google tokens Automation project (C+D, YouTube + Instagram ka ek hi Supabase) me** rehte hain. Workspace sirf gateway se usse baat karta hai. Office ya Team YouTube ko manage nahi karte; sirf Workspace admin (permission `automations.manage`) karta hai.

6.1 Ek baar ka setup (kahan kya daalna hai)

Step	Kahan	Kya
1	Supabase: **naya Automation project** (YouTube + Instagram dono ke liye)	Authentication me sign-ups OFF. SQL Editor me `LEXON_D_youtube.sql` phir `LEXON_C_instagram.sql` Run, phir AUTOMATION_PREFLIGHT.sql
2	Apne computer par `lexon-backends/youtube`	`supabase link --project-ref <D-ref>`, phir `supabase functions deploy youtube-oauth --no-verify-jwt`, `youtube-sync --no-verify-jwt`, `youtube-ai --no-verify-jwt`
3	Automation project → Edge Functions → Secrets	WORKSPACE_URL, OAUTH_STATE_SECRET, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, ANTHROPIC_API_KEY, AI_MODEL (optional)
4	Google Cloud Console	YouTube Data API v3 + YouTube Analytics API **Enable**. OAuth redirect URI: `https://<D-ref>.supabase.co/functions/v1/youtube-oauth`
5	**Workspace** → Integrations → Lexon Workspace → **Automation backend** → **Link project**	Is project ka URL + **service_role key** (sirf yahin; Vault me encrypted) → "Verify and link" (YouTube + Instagram dono ek saath)

Step	Kahan	Kya
6	Workspace → Integrations → YouTube → **Test connection**	"Connection healthy, schema complete" aana chahiye
7	Workspace → Automations → YouTube	Connect Google account → **Sync now**

6.2 Ek hi Supabase: YouTube + Instagram

- Workspace → Integrations → Lexon Workspace → **Automation backend (YouTube + Instagram)**: ek hi "Link project" (ek URL + ek service key). Key Vault me encrypted rehti hai.
- **All-or-nothing:** agar dono me se ek schema missing hai to kuch adhoora link nahi rehta. "Test connection" dono ko check karta hai; "Unlink" dono hatata hai.
- **Safety check:** us project me `supabase/AUTOMATION_PREFLIGHT.sql` chalao (sirf padhta hai). Har row ok = true: tables maujood, RLS ON, anon/authenticated ke liye koi policy ya privilege nahi.
- **Merge SQL:** ek combined SQL (LEXON_CD) tab banega jab LEXON_D_youtube.sql, youtube functions ka code aur aapka Instagram schema mil jaye. Tab tak haath se merge ke safe steps: docs/AUTOMATION_SHARED_PROJECT.md.
- **Trade-off:** ek key dono ka data kholti hai. Leak hui to dono expose; isliye sign-ups OFF, key sirf Vault me, kabhi commit nahi.

6.3 Roz ka management

- **Sync / analytics / video comparison / AI Studio:** Workspace → Automations → YouTube.
- **Gateway whitelist:** sirf tables youtube_videos, youtube_channel_snapshots, youtube_ai_outputs, youtube_connection aur functions youtube-oauth/-sync/-ai khulte hain. Workspace in tables me likh nahi sakta (read-only).
- **Execution history:** har sync/AI call Automations page par automation_logs me dikhti hai aur Activity feed me bhi.
- **Error:** "not linked" = step 5 baaki; "schema failed" = Automation project me SQL run karo; Google token error = Connect Google dobara.

YouTube ka SQL aur functions ka code maine nahi dekha (LEXON_D_youtube.sql aur lexon-backends/youtube is package me nahi the). Upar ke steps Workspace code (gateway/UI) aur purani MANUAL_SETUP se hain. Pehli baar run par chhote errors aa sakte hain.

7. Office aur central API Manager

Ek hi jagah API keys: Workspace → Integrations → Lexon Office → **"AI models and APIs"** (27 free/trial providers ya Custom API). Keys Supabase Vault me encrypted hoti hain, browser ko kabhi wapas nahi aatin. Office ke andar alag API Manager **nahi** hoga.

Kaun	Kya karta hai
Workspace admin	Key/model yahan add, test, remove karta hai (scope "office")
`office-ai` function (project A)	Office ke server ke liye gateway. `models`: connected models ki list (key ke bina). `chat`: key Vault se utha kar provider ko call karta hai, sirf text wapas deta hai
Office Workers (project B ka server)	`x-office-secret` header ke saath office-ai call karte hain. Key kabhi Office me store nahi hoti

Setup

1. Project A: secret `OFFICE_GATEWAY_SECRET` (lamba random) set karo; `supabase functions deploy office-ai --no-verify-jwt`.
2. Project B (Office): wahi value secret me; `WORKSPACE_FUNCTIONS_URL=https://<A-ref>.supabase.co/functions/v1`.
3. Office me jo duplicate API/keys section hai use hata do (UI, table, stored keys). Workers office-ai ko **Office ke server (Edge Function)** se call karein, browser se nahi.
4. Workspace me key daalte hi connected Workers use kar sakte hain: models list har baar office-ai se aati hai.

Office app ka code mujhe nahi mila, isliye Office ke andar ka duplicate section maine khud nahi hataya. Workspace ki taraf (gateway + Office page par read-only models card) tayyar hai. Office zip bhejoge to Workers ko office-ai se jod dunga. Snippet: docs/OFFICE_INTEGRATION.md.

8. Windows .exe aur Web: ek hi app, har entry par login

- **Ek hi code:** Windows app (Electron) sirf `dist` me bani web build ko dikhata hai. Alag code nahi. Workspace aur Team dono ke alag desktop shells hain (`lexon://` aur `lexon-team://`).
- **Web:** GitHub Pages / Cloudflare / Netlify par static build; `public/_headers` me strict CSP. Windows: GitHub Actions → "Build Windows app (.exe)" → artifact download.
- **Login mandatory:** har protected route `RequireAuth` ke andar hai. Bina session /#/messages jaisa direct link kholo to /login par bhej deta hai. Check fail ho (network/error) to **band** rehta hai, khulta nahi.
- **Asli suraksha server par:** koi JS file ya URL se UI khul bhi jaye, data bina valid JWT ke nahi milta (RLS). 2FA wale users ko aal2 chahiye. Deactivated user ko "Access removed".
- **Desktop shell:** page me Node nahi, sandbox + contextIsolation, sirf bundled app (`lexon://app`) load hota hai, file:// aur path traversal block, navigation lock, Google login system browser se aur deep link se wapas.

```
Windows build (Windows PC par):
cd apps/desktop
npm install
npm run dist:win        # output: apps/desktop/dist/*.exe
# ya GitHub: Actions -> "Build Windows app (.exe)" -> Run workflow
```

Unsigned .exe par Windows SmartScreen warning dikhayega (code-signing certificate chahiye). Maine .exe yahan (Linux) build nahi kiya; workflow aur config tayyar hain.

9. Kaun sa data kahan rahega

Data	Kahan	Kaun dekh sakta hai
Login, password hash, 2FA factors	Supabase Auth (project A, auth schema)	Koi app nahi; sirf Supabase
Profile (naam, photo, about), presence	A: profiles	Workspace members
Phone, email, address, emergency contact	A: profile_private	Khud + Admin
Roles / permissions / approved emails	A: workspace_members, role_permissions, authorized_emails	Admin (member sirf apna role)

Data	Kahan	Kaun dekh sakta hai
Projects, tasks, attendance	A: projects, tasks, task_assignees, attendance	Admin sab; member sirf apne tasks
Messages, calls	A: conversations, messages, call_sessions	Sirf chat ke members / call ke 2 log
Files, avatars	A: files + Storage (workspace-files, avatars)	Workspace members (avatars public read)
Notifications, activity, login history	A: notifications, activity_logs, security_events	Khud; Admin activity sab
API keys, Google/GitHub tokens, linked project keys	A: Vault (integration_secrets). Client policy nahi	Sirf Edge Functions (service role)
Office members/agents/layout	Project B	Office app
YouTube videos, analytics, Google token	Automation project (C+D)	Sirf Workspace admin, gateway se (read-only)
Instagram rules/events/messages	Automation project (C+D), baad me	Sirf Workspace admin, gateway se

10. Instagram: abhi sirf structure

- Tayyar: Automations page par Instagram card, tables `automations`/`automation_logs`, permission `automations.manage`, gateway me `instagram` project (tables ig_*, sirf ig_rules writable), Integrations me "Link project", Vault me tokens.
- Aapki Instagram guide aane ke baad: project C ka SQL reconcile, webhook Edge Function, rules editor. Plan: `docs/INSTAGRAM_INTEGRATION.md`.
- Meta developer app, professional account aur messaging permissions ke liye Meta app review chahiye.

11. Uploaded SQL aur PDF ki jaanch: Workspace vs Team

Check	Result
LEXON_A_workspace_team.sql (aapki file) vs Workspace ka `ALL_IN_ONE_A.sql`	**Bilkul identical** (line-by-line diff khali, 1295 lines). Yani migrations 0001-0008
Team zip me alag SQL?	**Nahi.** Team README: "uses the same Supabase project as Workspace"
Team ke tables / RPCs	17 tables (profiles, tasks, messages, notifications, call_sessions, files ...) aur 5 RPCs (create_group, my_conversations, set_security_retention, start_direct, toggle_reaction): **sab file me maujood**
Team zip ka PDF vs Workspace ka PDF	**Byte-identical** (same hash)
PDF vs asli SQL	PDF section 7.2 bolta hai "migrations 0001 to 0007"; asli file me **0001-0008** hai. PDF purana hai. PDF ye bhi kehta hai ki SQL abhi kisi Supabase par run nahi hua tha

Result: Workspace aur Team ke SQL/schema me **koi difference nahi**, kyunki Team ka apna schema hai hi nahi. Same karne ko kuch baaki nahi. Jo badla wo code me hai:

- Workspace me sirf Google login tha, Team me sirf email code. **Dono me ab Google + password** (Team me email code bhi).
- Team ka Supabase client ab Workspace ki tarah pkce flow use karta hai (desktop Google login ke liye zaroori).
- Team ko Workspace ka Modal, Logo, sidebar/header style mila; halke rounded corners aur iPhone-style bottom bar.
- Dono RequireAuth ab active membership bhi check karte hain. Naya SQL sirf 0010 (chat access) aur 0009 (repair) joda hai; dono re-runnable.

12. Setup checklist

#	Karo	Ho gaya?
1	Supabase project A banao; URL + anon key; SQL file run; DIAGNOSE.sql me sab ok	
2	Auth: Email ON, min password 8, Google ON, redirect URLs (lexon://app/**, lexon-team://app/**), SMTP, TOTP	
3	Edge Functions deploy + secrets (ALLOWED_ORIGINS, APP_URL, TEAM_APP_URL, OAUTH_STATE_SECRET, OFFICE_GATEWAY_SECRET)	
4	Workspace me Google se pehle login (aap Owner)	
5	Settings → Security: password set, 2FA ON	
6	Team → Invite; member Team app me code se login, phir password set	
7	Unapproved Gmail se Google login: "Access denied" aana chahiye	
8	Direct /#/messages bina login: /login par jana chahiye	
9	Integrations → Lexon Office → AI key add; Office page par model dikhe	
10	Automation project setup (section 6); AUTOMATION_PREFLIGHT.sql sab ok; Workspace me link + Test connection	
11	Windows: Actions se .exe banao, install, Google + password login test	
12	Dono apps deploy (GitHub Pages secrets: VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY)	

Troubleshooting

Dikkat	Wajah / hal
Home "Could not load data"	SQL migrations adhoori. DIAGNOSE.sql chalao, phir 0008, 0009. Ab Home aadhi load bhi ho to dikhata hai aur peela notice batata hai kya fail hua
"Your account is not linked to a workspace"	Account workspace_members me nahi ya active=false. DIAGNOSE.sql ki aakhri query dekho
Google login par "Access denied"	Gmail authorized_emails me nahi. Admin approve kare / Team → Invite
Password login "Invalid email or password"	Password abhi set nahi hua (pehle Google/code se login karke Settings me set karo) ya galat hai
Edge function "Failed to send a request" / CORS	Function deploy nahi hui ya ALLOWED_ORIGINS me URL nahi
Desktop Google login wapas nahi aata	Redirect URLs me lexon://app/** (Team: lexon-team://app/**) nahi hai

13. Jo abhi test nahi hua (seedhi baat)

- Ye SQL maine kisi live Supabase par run nahi kiya (mere paas aapka project nahi hai). Pehli run par chhoti dikkat aa sakti hai; DIAGNOSE.sql se pakdo.
- Windows .exe yahan build nahi hua. Web build aur TypeScript check dono apps ke pass hue; Login pages dono apps ki browser me check hui (redirect, Google button, password form, galat password ka message).
- Google/password login asli Supabase + Google Cloud settings ke saath aap pehli baar chalaoge: section 5.4 ka har step zaroori hai.
- YouTube + Instagram ke ek combined SQL ke liye LEXON_D/LEXON_C aur youtube functions ka code chahiye (abhi sirf safe merge steps + preflight diya hai). Office app aur project B ka code bhi mujhe nahi mila. Office ke duplicate API section ko hatane ka kaam Office zip milne par.
- Password ki exact length "8": app minimum 8 (max 72) maanti hai. Agar exactly 8 chahiye to batao.